

15. The interrupt vector for INT 13H is stored in memory at:

- A. 0012H B. 0048H C. 004CH D. 0084H

16. The interrupt generated by the execution of an interrupt instruction INT is

- A. External interrupt B. Software interrupt C. NMI interrupt D. Interrupt in interrupt

17. At the end of interrupt when the CPU executes IRET,

- A. clears the trap flag B. Contents of IP and CS are retrieved
C. clears the interrupt flag D. The control transfers TO ISR to main program

18. The equivalent instructions of STOSB assuming DF flag = 0:

- A. MOV [DI], AL; INC DI B. MOV [SI], AL; INC DI
C. MOV AL, [SI]; INC SI D. MOV [DI], AL; DEC DI

19. The values of CX and AL after executing the following code are:

```
MOV AL, 0FFH
MOV CX, 4
L1: ADD AL, 1
LOOPZ L1
```

A. CX=3; AL=1

B. CX=1; AL=2

C. CX=3; AL=0

D. CX=2; AL=1

20. Determine the value of register AL after execution of the following assembler codes:

```
SI → X DB '123456'
CLD
LEA SI, X
LODSB
MOV BL, AL
LODSB
XOR AL, BL
```

A. AL=03H

B. AL=32H

C. AL=02H

D. AL=33H

21. Find the content of register AL after execution of the following code:

```
PI → X DB 'ABCD*KLM+A123'
CLD
MOV CX, 14
MOV AL, '*'
LEA DI, X
REPNE SCASB
MOV SI, DI
LODSB
```

A. AL='3'

B. AL='*'

C. AL='D'

D. AL='k'

22. What will be appears on the screen after executing the following code:

```
STR DB "KLMNC", 0DH, "ABmk", '$'
MOV AH, 9
MOV DX, OFFSET STR
INT 21H
```

A. ABmkC

B. KLMAB

C. ABmkKLmk

D. KLMNC

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11
A	D	B	D	A	B	B	D	A	C	C
Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20	Q21	Q22
B	A	C	BC	B	B	A	AD	A	D	A

1. The value of AL and carry flag (CF) after the following instructions have executed is

A. AL=22; CF=0

C. AL=88; CF=1

B. AL=22; CF=1

D. AL=88; CF=0

2. The offset of a particular segment varies from _____ to _____

A. 0000h to FFFh

B. 0000h to FFFFh

C. 00h to FFh

D. 00000h to FFFFFh

3. In 8086 microprocessor the overflow flag is set when:

A. The result is more than 16 bit

B. If CF and SF are set to 1

B. Signed number goes out of their rang after an arithmetic operation

D. During subtraction

4. During the execution of an interrupt the data pushed into stack is the contents of:

A. IP

B. CS

C. flag register

D. all of them

5. An interrupt breaks the execution of program and diverts it to execution to:

A. Interrupt service routine

B. execution unit

C. counter word register

D. control unit

6. After execution of the following code register AL contains

MOV AL, 'a'

ADD AL, 2

AND AL, 1101 1111b

a) AL='c'

b) AL='C'

c) AL='a'

d) AL='A'

7. If 1111 1011 is the 8-bit 2's complement representation of a number N, what is the decimal value of N?

A. -11

B. -5

C. 10

D. 12

8. 18. Number of the times the instruction sequence below will loop before coming out of loop is

MOV AL, FFH

L: ADD AL, 1

JNZ L

A. 1

B. 4

C. 3

D. 0

9. Find the content of AL and zero flag (ZF) and carry flag (SF) after execution of the following code:

MOV AL, 0FDH

ADD AL, 3

ADC AL, 0FFH

A. AL=FFH, ZF=0; SF=1

B. AL=00H, ZF=0; SF=1

C. AL=FFH, ZF=0; SF=0

C. AL=02H, ZF=0; SF=1

10. If offset is a 12-bit number address, then the maximum possible locations are _____

A. 2 KB

B. 1 KB

C. 4 KB

D. 1 MB

11. In the following instruction sequence, show the value of AL:

MOV AL, 0FAH

NOT AL

XOR AL, FFH

A. AL=FBH

B. AL= 05H

C. AL= FAH

D. 06H

12. All the functions of the ports of 8255 are achieved by programming the bits of an internal register called

A. Address register

B. read logic control

C. Control word register

D. data bus control

13. How many bits of data can be transferred between the 8255 PPI and the interfaced device at time?

A. 16 BITS

B. 24 BITS

C. 8 BITS

D. 20 BITS

14. what will be stored in Y after executing the following codes

S) X DB "123"

D) Y DB "ABCDE"

MOV SI, OFFSET X

MOV DI, OFFSET Y+1

MOV CX, 2

REP MOVSB

A. 123DE

B. 123

C. A12CDE

D. AB12E

SI → DI

QUESTION #4 (5 points)

Fill in the blank spaces with the correct missing components of an 8086 assembly program to accomplish the following

This section of assembly code clears the entire screen and sets the foreground color to white and the background color to red.

MOV AH, 06

MOV AL, 00

MOV CX, 0

MOV DX, 184FH

MOV BH, 4Fh

INT 10

; This section of assembly code sets the cursor position to the coordinates (x, y) = (39, 12).

MOV AH, 02

MOV BH, 0

MOV DL, 39

MOV DH, 12

INT 10h

; This section of assembly code prints \$-terminated string array named string1

MOV AH, 09

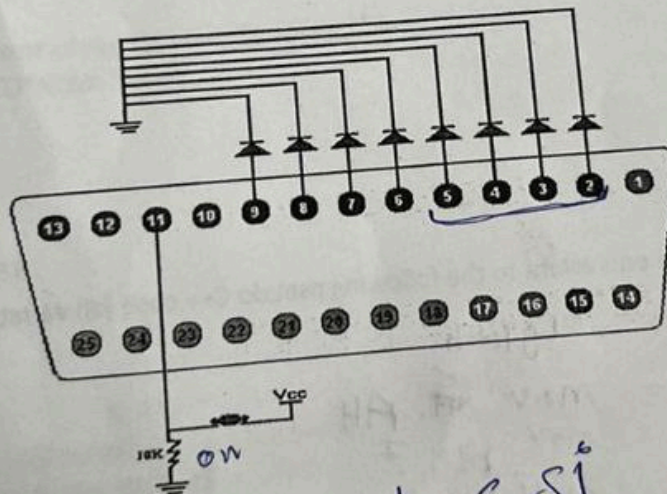
MOV DX, offset string1

INT 21h

HLT

String1 DB 'ABCDEF\$'

Question # 5: Based on the figure below, which illustrates eight LEDs connected to the parallel port pins P2-P9 associated with the data register (378h), and one switch connected to the parallel port pin (P11) associated with the seventh bit (S6) of the status register (379h), write an assembly program that turns on the lower four LEDs when the switch is pressed. Conversely, the upper four LEDs should be turned on when the switch is not pressed. [5pts]



```
org 100h
mov SI, 0
mov CX, 10
mov DX, 378h
mov AL, DATA[SI]
out DX, AL
CALL DELAY
```

```
inc SI
loop L1
jmp SJ
DAT DB, 1, 2, 3, 4, 5, 6, 7, 8
```

QUESTION#2: Find the outputs of the following program segments (10 POINTS)

<pre> X DD 1345700CH MOV DX,X MOV BX,200FH PUSH BX PUSH DX TEST DL,BH POP DX </pre> DX 700C BX= 200F;	<pre> SI X DW 0'KLM+ABC' DI Y DW 0'KLM+WBD' CLD MOV SI, OFFSET X MOV DI, OFFSET Y MOV CX, 100 REPE CMPSW MOV AX,[SI] MOV DX,[DI] PUSH AX PUSH DX POP BX </pre> AX= A13 ; BX= WB CX= 98 DX= WB
<pre> MOV CX,10 MOV AX,0 L:ADD AX,CX CMP AX,33 LOOPNZ L </pre> CX= 5 AX= 33	<pre> MOV AL,FEH MOV BL,0 MOV CL,4 ADD AL,2 RCR AL,CL ADC BL,3 </pre> AL= 80h ; BL= 03h

QUESTION #3(8 POINTS)

- Write instructions sequence equivalent

a) LOOP instruction

~~DEC CX~~ DEC CX
~~JNZ L~~

b) NEG AX

MOV AX,5
NEG AX

c) MOVSB

MOV DI, SI
INC DI
INC SI

d) Write an 8086 assembler code that is equivalent to the following pseudo C++ code (all variables are unsigned 16-bit integers). [4pts]

if (a1 <= b1) AND (b2 > a2)
X = 5;
else
X = 4;

org 100h
mov AX, AH
mov BL, Z
mov AL, Y
mov b2, F
mov a2, E
CMP al, bl
JGL
CMP b2, a2
JLL AH
MOV AX, 5
~~MOV~~ J ENCL
LI: MOV AX, 4
HLE

ZDB 00
YDB 10
FDB 11
EDB 01